

LAB 1: REQUIREMENTS ENGINEERING

Personalized Meal & Diet Subscription Manager

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Problem Statement: #19

Domain: Healthcare & Telemedicine

1. Requirements Table

The following functional and non-functional requirements are derived from the given problem statement. Each requirement is written using a clear “The system shall...” format and includes measurable acceptance criteria.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments	
FR-001	Functional	The system shall generate weekly meal plans strictly adhering to the subscriber's metabolic constraints, calorie goals, and allergen exclusion list.	HIGH	Pass: Generated weekly menu contains zero excluded allergens and stays within the specified calorie and dietary limits. Fail: Menu contains an excluded allergen or exceeds the specified limits.	Ensures that meal plans are personalized and safe for the subscriber.	Clearly defines the dietary constraints and allergen requirements.	
FR-002	Functional	The system shall allow a subscriber to view their personalized weekly meal plan, including	HIGH	Pass: Subscriber can open the weekly plan and view meals assigned for	Allows subscribers to clearly understand their	Specify that the weekly plan should display all seven days.	

		the meals scheduled for each day.		all seven days. Fail: Any scheduled day or meal is missing from the displayed plan.	assigned meals.	
FR-003	Functional	The system shall allow a dietitian to update a subscriber's dietary constraints, including allergies, diabetic limits, and calorie goals.	HIGH	Pass: After the dietitian saves a changed constraint, the updated value is displayed in the subscriber's dietary profile. Fail: The old value remains after saving.	Keeps the subscriber's dietary information current for meal planning.	Clearly identifies the dietary information that the dietitian can update.
FR-004	Functional	The system shall allow the subscriber to modify an upcoming meal delivery schedule by pausing, skipping, or rescheduling a delivery.	HIGH	Pass: A valid pause, skip, or reschedule request is saved and the updated delivery schedule is displayed. Fail: The requested change is not reflected in the schedule.	Provides flexibility when the subscriber's plans change.	Specify that modifications are allowed only before the dispatch deadline.
FR-005	Functional	The system shall track the daily fulfillment status of the subscriber's meal deliveries.	MEDIUM	Pass: Each scheduled delivery is recorded with its current fulfillment	Helps monitor whether scheduled meals are being fulfilled.	Specify the fulfillment statuses, such as pending, prepared, dispatched, and delivered.

				status. Fail: A scheduled delivery has no recorded fulfillment status.		
NFR-001	NonFunctional	The system shall support schedule modifications such as pause, skip, and reschedule up to 12 hours before dispatch	HIGH	Pass: A valid schedule modification made at least 12 hours before dispatch is accepted and recorded. Fail: A valid modification made within the allowed period is incorrectly rejected.	Provides flexibility while allowing sufficient time for delivery preparation.	The 12-hour modification limit is clearly measurable and testable.
NFR-002	NonFunctional	The system shall protect subscriber dietary and health-related information from unauthorized access.	HIGH	Pass: An unauthorized user attempting to access subscriber dietary information is denied access. Fail: An unauthorized user can view or modify the information.	Protects sensitive subscriber information and prevents unauthorized access.	Specify that only authorized users such as the subscriber and dietitian can access the information.

2. Use-Case Flow Specification

Use Case: Generate Weekly Meal Plan (UC-02)

Preconditions

1. The subscriber has a registered account in the system.
2. The subscriber's dietary profile contains the required metabolic constraints, calorie goals, and allergen information.
3. The system has access to the meal and dietary information required to generate the plan.

Postconditions

1. A personalized weekly meal plan is generated for the subscriber.
2. The generated plan follows the subscriber's calorie and dietary constraints.
3. The weekly meal plan is stored and made available for the subscriber to view.

Main Success Scenario

1. The subscriber logs into the Personalized Meal & Diet Subscription Manager.
2. The subscriber selects the option to generate a weekly meal plan.
3. The system retrieves the subscriber's dietary profile.
4. The system checks the subscriber's metabolic constraints, calorie goals, and allergen exclusion list.
5. The system generates a weekly meal plan according to the specified constraints.
6. The system verifies that the generated meals do not contain excluded allergens and follow the specified dietary limits.
7. The system saves the generated weekly meal plan.
8. The system displays the completed weekly meal plan to the subscriber.
9. The use case ends successfully.

Alternate Flow – Invalid or Conflicting Dietary Constraints

1. The system detects that the subscriber's dietary constraints are incomplete or conflicting.
2. The system displays a message asking the subscriber to update the required dietary information.
3. The subscriber updates the dietary profile.

4. The system validates the updated information and generates the weekly meal plan.
5. If the information is still invalid, the system does not generate the meal plan and informs the subscriber to contact the dietitian.

3. Actors and Use Cases

Actors

1. **Subscriber** – Views personalized meal plans and manages meal delivery schedules.
2. **Dietitian** – Manages and updates the subscriber's dietary constraints and meal planning information.
3. **Delivery Service** – Tracks the fulfillment status of scheduled meal deliveries.

Use Cases

1. **UC-01: Manage Dietary Profile**
2. **UC-02: Generate Weekly Meal Plan**
3. **UC-03: View Weekly Meal Plan**
4. **UC-04: Manage Delivery Schedule**
5. **UC-05: Track Delivery Fulfillment**